

Invariant normalized kernel over an infinite field

$$\begin{aligned} H &= qD, \quad K = \ker H, \quad A(K) \subseteq K, \quad A_\tau = A + \tau H \\ \tau &\in \operatorname{Hom}(\operatorname{im} H, R), \quad \{\tau H\} = \{T \in \operatorname{End}(R) : T|_K = 0\} \end{aligned}$$



Choose a vector-space complement $R = K \oplus X$

$$\mathcal{O}_{\text{tilt}}(A) = \{B : B|_K = A|_K\}, \quad B = \begin{pmatrix} A|_K & C \\ 0 & B_X \end{pmatrix}$$

Both $C \in \operatorname{Hom}(X, K)$ and $B_X \in \operatorname{End}(X)$ are arbitrary.



Two block-diagonal witnesses, $c \neq d$ and $r = \dim X > 0$

$$\begin{aligned} B_c &= A|_K \oplus cI_X, & B_d &= A|_K \oplus dI_X \\ \chi_{B_c} &= \chi_{A|_K}(S - c)^r, & \chi_{B_d} &= \chi_{A|_K}(S - d)^r \\ \mu_{B_c} &= \operatorname{lcm}(\mu_{A|_K}, S - c), & \mu_{B_d} &= \operatorname{lcm}(\mu_{A|_K}, S - d) \end{aligned}$$



Common divisors over the full tilt family

$$\gcd_{\tau} \chi_{A_\tau} = \chi_{A|_K}, \quad \gcd_{\tau} \mu_{A_\tau} = \mu_{A|_K}$$

If $X = 0$, the orbit is a singleton. Zero-dimensional polynomials equal 1.
A more restricted tilt family can have a larger gcd.